SEARCH SKILLS

- For google: "exact phrase" Double quotes indicate that you are looking for pages with the exact word or phrase.

site: This operator lets you specify the domain name to which you want to limit your search. For ex: we can search for success stories on thm using site: tryhack.com success stories

- : You might be intersted in learning about pyramids but you don't want to view tourism websites. pyramids -tourism or -tourism pyramids

filetype: If you want to find cybersecurity presentations, try searching for filetype: ppt cybersecurity

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• Specialized search engines:

1= Shodan: It allows you to search for specific type and versions of server, networking equipments, industrial control system and IoT devices, You may want to see how many servers are still running Apache 2.4.1 and the distribution across countries. For this search apache 2.4.1, it will return list of server with string "apache 2.4.1" in their headers

2= Censys: It focuses on Internet connected devices and systems, such as servers, ~~re~~ hosts, websites, certificates and other internet assets.

3= Virus Total: Virus Total is a website that allows users to upload files or provide URL's to scan them against numerous antivirus engines and website scanners in a single operation

4= Have I been Pwned: It tells you if an email address has appeared in a leaked data breach. Finding ones' email in a leaked data breach indicates leaked private information

• CVE (Common Vulnerabilities and Exposures): A CVE is a publicly known security vulnerability that has been: Identified, Verified, Given a unique id.
This MITRE Coorporation maintains the CVE system.

• Exploit database: It is a website which lists that collects and exploit codes written by various security proffessionals. These exploit codes show how attackers can take advantage of vulnerabilities in system.

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In simple terms: Hackers working computer

- Attack box: An attack box is a machine (usually linux) that an attacker or security tester uses to perform attacks, scans, and testing on target system.

ACTIVE DIRECTORIES BASIC:

- Windows domains: A windows domain is a centralized way to manage many computers and users from one place.

Without a domains:	With a domain
Each computer has it own users	One login works everywhere
No central power	Admin controls everything.

- Active directory: It is a directory service that stores and manages everything in a windows domain.

- Domain Controller (DC): A DC is a windows server that runs active directory.

It acts as a catalouge that holds the information of all the "objects" that exist on your network such as users, groups, machines, printers and many others.

a) Users: Users are one of the objects known as security principal. You can say that a security principal is an object that can act upon resources in the network like printers and files.

b) Machines: They are another types of objects, for every computer that joins the active directory domain, a machine will be created. They are also security principals.

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c) Security groups: They are also considered as security principals
They can have both ~~computer~~ users and machines as members
Several security groups are created by default:

Domain Admins: users in this have administrative power over domain
Server Operators: Users in this group administer DC.
Backup operators: Users in this group can access any file, ignoring their permissions. They are used to perform backup on the computer.
Account Operators: Users in this can modify or create accounts
Domain Users: Includes all existing users in the domain.
Domain Computers: " " " computers " "
Domain Controllers: " " " DC in the domain

- Organization unit (OU): An OU is a containers in Active directory used to organize users, computers and groups.
They are used to:
Organize objects (by department, location, role)
Apply group policies

Security groups and OU difference? Like we can organize objects of the AD by putting them in different groups also?
OU are used for applying policies to users and computer while on other hand security groups are used to grant permission over resource. A user can be part of multiple S.G. but can only be a part of single OU.

- Delegation: This gives some specific users some control over some OU's. This process is known as delegation and allows you to grant users specifics privileges to perform advanced tasks on OU's without needing a domain administrator.

How to do this? Right click on OU and select delegate control.

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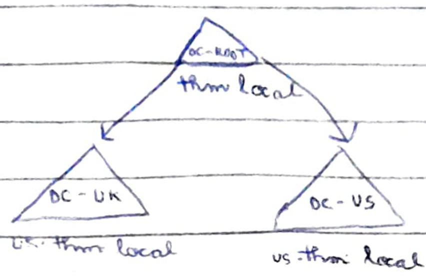
- GPO (Group Policies Object): It is used to deploy different policies for each OU individually.
GPO are generally a collection of settings that can be applied to OU. GPO contains policies either aimed for users or computers. To configure GPOS, you can use Group policy Management tool.

GPO distribution: GPO are distributed over a network via network share called $SYSVOL, which is stored in the DC. All the users in the domain have access to this share over the network to sync their GPOs periodically.

- Authentication methods: whenever a user tries to login to a service using domain credentials, the service asks the domain controller if they are correct. Two protocols are used for network authentication in windows domain:
i) Kerberos: used by any recent version of windows. This is default protocol
ii) NetNTLM: Legacy authentication protocol kept for compatibility purposes.

Kerberos uses tickets to authenticate.
NetNTLM is less secure than Kerberos. Uses challenge-response process to authenticate.

- Trees: Suppose my companies grows from one country to another. Now here trees comes into play.
Trees are a collection of domain that share the same namespace.



- Forests : The union of several trees with different namesp-ces into the same network is called forest.

- Enterprise Admine Group : The users under this group are very powerful they can control the entire forest.
a) Create / Delete domains
b) Modify forest wide settings
c) Controls all trees and Domain

- Trust Relationships : ~~This allows~~ Trust relationships b/w two domains allows user from a domain to access the resources from other domain.
If Domain AAA trust Domain BBB this means a user of BBB can be authorised to access resources on AAA. This is called one way trust relationships.